Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Zheng, Chiang, and Sheng et al. (2023)

Hins Fang Charles Costanzo

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$\rightarrow$ **LLM-as-a-Judge**



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→ LLM-as-a-Judge

Tried before, but never studied systematically.



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- (and how do we even **measure** that?)

Meet the models: Camelids in the Hot Seat



LLaMA-13B (Meta, 2023)



Vicuna-13B (LMSYS, 2023)

- These two are just our example. The paper evaluates up to 15 chatbots, including GPT-4 and Claude.

Photos: PA (llama) and Marshallhenrie (vicuña), Wikimedia Commons, CC BY-SA 4.0

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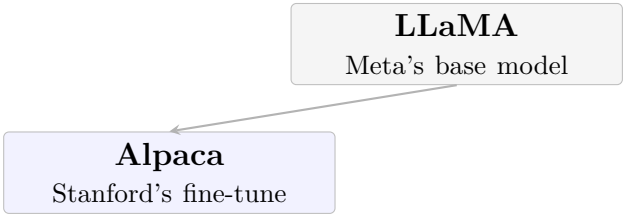
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LLaMA
Meta's base model

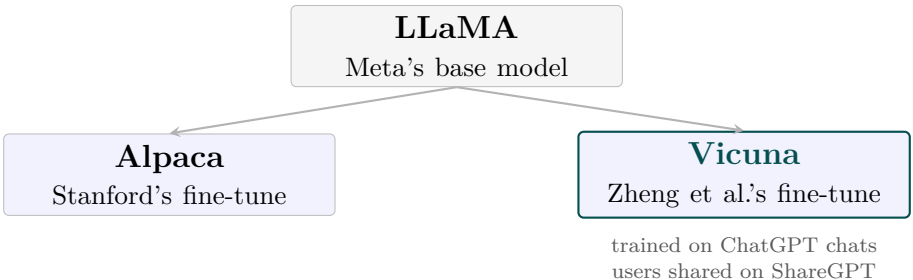
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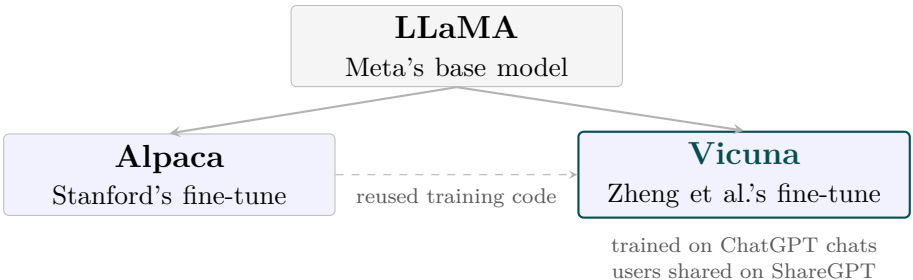
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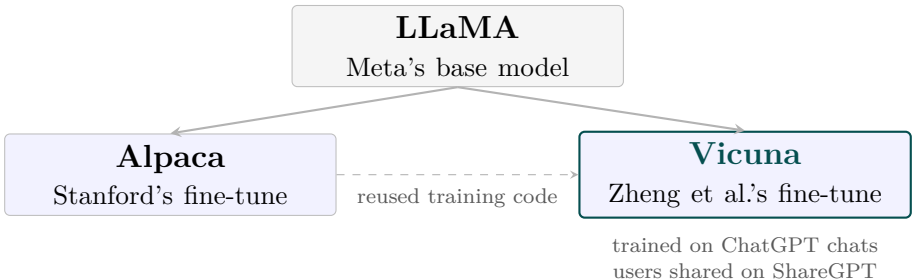
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Both are LLaMA's camelid descendants.

Meet the judges: LLMs on the Bench

GPT-4



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- All three are checked against **human** judges:
  - 58 experts (MT-Bench) and crowdsourced ratings (Chatbot Arena)
- The judges are also among the models evaluated, so they sometimes grade their own answers.

Example 1:

Question

If the FED buys bonds in the secondary market

(A) the money supply will increase.
(B) the money supply will decrease.
(C) the money supply will not be affected.
(D) the discount rate would be affected.

Output the answer only.

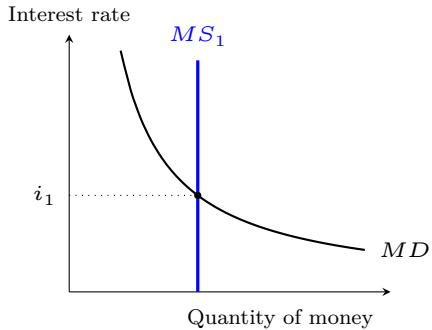
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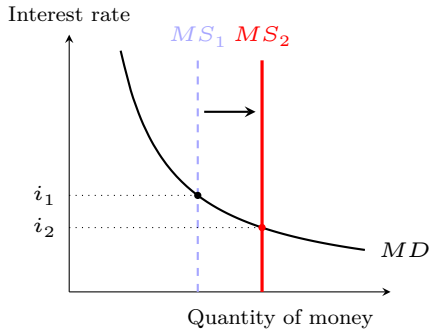
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Example 1

LLaMA-13B

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The Federal Reserve buys bonds in the secondary
market to increase the money supply.
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Vicuna-13B

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(A) the money supply will increase.
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Example 1 - Follow-up:

Follow-up Question

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How does it affect my daily life? Give 3 examples
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Follow-up Question

How does it affect my daily life? Give 3 examples

LLaMA-13B

The Federal Reserve buys bonds in the secondary market to increase the money supply. This affects your daily life by:

1. Increasing the money supply.
2. Increasing the money supply and lowering the discount rate.
3. Increasing the money supply and lowering the discount rate and increasing the money supply.

Vicuna-13B

When the Federal Reserve buys bonds in the secondary market, it can have several effects on the economy that can impact your daily life. Here are three examples:

1. Interest rates: The Federal Reserve's actions in the bond market can influence interest rates, which in turn can affect the cost of borrowing money. For example, if the Federal Reserve buys bonds and drives up demand for them, interest rates on things like mortgages, car loans, and credit cards may go up, making it more expensive to borrow money.
2. Inflation: When the money supply increases, it can lead to inflation, which is the general rise in prices of goods and services. This can impact your daily life by making it more expensive to buy things like groceries, gas, and other necessities.
3. Employment: The Federal Reserve's actions in the bond market can also have an impact on employment. For example, if the Federal Reserve's actions lead to economic growth and increased demand for goods and services, it can create more job opportunities and boost employment.

You be the judge



menti.com code **1479 0094**

Which assistant did better in this conversation?

- LLaMA-13B
- Vicuna-13B
- Tie

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Then we'll see what GPT-4 said.

Example 1 - GPT-4 as judge

GPT-4 Judgment

LLaMA-13B provided an incorrect response to the user's question about how the Federal Reserve buying bonds in the secondary market affects daily life. The answer given is repetitive and lacks clear examples of how the action impacts daily life.

On the other hand, Vicuna-13B provided a relevant and accurate response to the user's question about the Federal Reserve buying bonds. The answer includes three clear examples of how the action impacts daily life, such as interest rates, inflation, and employment.

LLaMA-13B should improve by providing relevant, clear, and accurate examples to address the user's question. **Vicuna-13B's response is more helpful, relevant, and detailed.**

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Vicuna more helpful, relevant, and detailed

- LLaMA: repetitive, no real examples
- Vicuna: three clear, accurate examples

Look back at turn 1

LLaMA-13B

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(A) the money supply will increase.

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The prompt said to "output the answer only."

- LLaMA provided its own answer.
- Vicuna outputted the correct answer verbatim from the four options listed.

Scores under different benchmarks

Example 1 is an MMLU[1] question.

Model	MMLU (5-shot)	MT-Bench score (GPT-4)
LLaMA-13B	47.0	2.61
Vicuna-13B	52.1	6.39

Table 8 in the paper. MT-Bench is scored 1–10 by GPT-4.

[1] Massive Multitask Language Understanding (Hendrycks et al., 2021)

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- On MMLU: a 5-point gap
- On human-style preference: Vicuna scores **2.4×** higher

MMLU was not designed to measure human preferences.

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What we'll dig into

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  - **MT-Bench**: 80 multi-turn questions, 3K expert votes
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- **Limitations:** LLM biases and reasoning errors, and addressing some of them
- **Agreement:** how often LLM judges match human preferences

References

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2021. URL https://arxiv.org/abs/2009.03300.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *NeurIPS 2023 Datasets and Benchmarks Track*, 2023. URL https://arxiv.org/abs/2306.05685.